

PHOENIX – Threat Analysis Integration Workflow Diagram

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Deliverable: Threat Analysis Integration – Integration Workflow Diagram

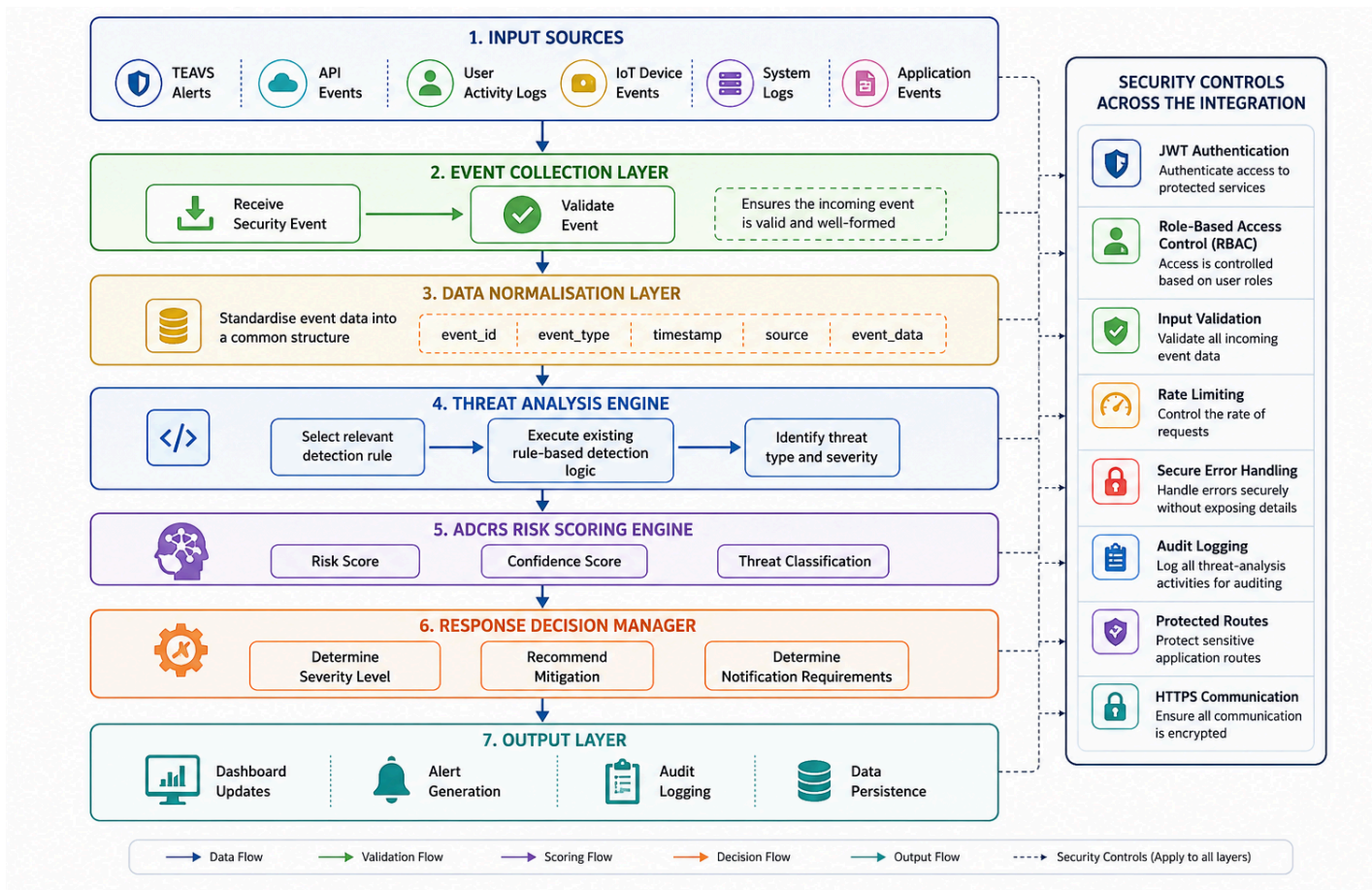
1. Purpose

The purpose of this workflow diagram is to provide a clear end-to-end representation of how security events are processed through the PHOENIX Threat Analysis Integration.

The workflow illustrates how security events from different PHOENIX sources are collected, validated and normalised before being processed by the Threat Analysis Engine and ADCRS Risk Scoring Engine. The resulting threat-analysis information is then used by the Response Decision Manager to determine the appropriate severity, mitigation recommendation and notification requirements before the final results are presented, stored and logged.

The workflow was developed to align with the existing Threat Analysis Integration Design Document and provide a clear technical reference for subsequent implementation, sequence modelling and integration testing.

2. Threat Analysis Integration Workflow



3. Security Controls Across the Integration

The following security controls apply across the Threat Analysis Integration workflow:

JWT Authentication

→ Authenticated access to protected integration functionality.

Role-Based Access Control (RBAC)

→ Access is controlled according to authorised roles.

Input Validation

→ Incoming event data is validated before further processing.

Rate Limiting

→ Supports control of excessive requests.

Secure Error Handling

→ Errors are handled securely within the integration.

Audit Logging

→ Threat-analysis activities and results can be recorded for auditing.

Protected Routes

→ Relevant application routes are protected from unauthorised access.

HTTPS Communication

→ Communication between relevant platform components is performed securely.

4. Example Event Flow

The following example demonstrates how a security event can move through the workflow.

Input Event

Event ID: EVT-001
Event Type: login_activity
Source: TEAVS
Timestamp: 2026-08-03T06:30:00Z

Processing

TEAVS Event
↓
Event Collection & Validation
↓
Data Normalisation
↓
Threat Analysis Engine
↓
ADCRS Risk Scoring
↓
Response Decision Manager
↓
Output Layer

Example Output

Event ID: EVT-001
Threat Type: Login Attack
Severity: High
Risk Score: 82
Confidence Score: 0.91
Recommended Action: lock_account
Status: detected

The final result can then be displayed through the dashboard, used for alert generation, stored for persistence and recorded through audit logging.

5. Workflow Summary

The Threat Analysis Integration workflow provides a structured process for handling cybersecurity events within PHOENIX.

Security events originating from TEAVS and other supported platform sources are first collected and validated. The events are then normalised into a common structure before being evaluated using the existing rule-based Threat Analysis Engine.

The event information is subsequently forwarded to the ADCRS Risk Scoring Engine, which provides a risk score, confidence score and threat classification. These results are combined and processed by the Response Decision Manager to determine the severity level, recommended mitigation and notification requirements.

Finally, the processed threat-analysis result is passed to the Output Layer, supporting dashboard updates, alert generation, audit logging and data persistence.

This workflow provides a visual and technical foundation for the subsequent **Threat Analysis Sequence Diagram** and **Integration Testing Plan**.